

We calculate the cross correlations of the order parameters, which take the form:

$$\begin{aligned}
g_{pm}(\vec{r}_1, \vec{r}_2) &= \langle \delta m(\vec{r}_1) \delta p(\vec{r}_2) \rangle \\
&= \sum_{\mathbf{k}} \langle \delta m_{1\mathbf{k}} \delta p_{1\mathbf{k}} + \delta m_{2\mathbf{k}} \delta p_{2\mathbf{k}} \rangle e^{i\vec{k} \cdot \vec{r}} \\
&= \frac{1}{2\pi^2} \cdot \frac{k_B T}{C_m \gamma_{me} r} \int_0^{k_{max}} k \sin(kr) dk \\
&= \frac{k_B T}{2\pi^2 C_m \gamma_{me} r^3} (\sin(k_{max} r) - r k_{max} \cos(k_{max} r)).
\end{aligned} \tag{62}$$

We see that the correlation length ξ_p near and above the magnetic phase transition is enhanced, whereas the cross correlations exhibit an oscillatory behavior and evolve continuously through T_{RM} .

V. CONCLUSIONS

In this paper we examined a simple Landau approach for a system with two single-component order parameters representing ferroelectricity and ferromagnetism, respectively. The obtained results are consistent with experiment. Magnetoelectric coupling introduces a strong renormalization of the ferromagnetic transition temperature (enhancing it by a factor of 50) causing the magnetic phase transition to be observed at $T_{RM} = 100.5K$. A number of coupling effects such as the enhancement of electric polarization, the anomaly in the dielectric susceptibility, and a fairly large negative magnetocapacitance, occur concomitantly in the vicinity of T_{RM} . We introduced a simple extension of the Arrot plot which we called in text *Arrot planes*. We consider this concept to be potential useful in evaluation of the phase transition temperature while using only field dependence of unbounded with it order parameter. We have also estimated the contribution of the Gaussian fluctuations to the specific heat and have noted that short-range-order effects are not accounted properly. It would be important to understand the microscopic reasons of such a strong enhancement of the magnetic exchange interactions caused by a monoclinic lattice distortion which leads to the appearance of the ferroelectric dipole moments.